



Stefano Giacomelli

Student ID No.: 295787

Ph.D. in ICT - System Engineering, Telecommunications and HW/SW platforms

Department of Information Engineering, Computer Science, and Mathematics
(DISIM)

University of L'Aquila (UnivAQ), L'Aquila (Italy)

Latest update: 16 June 2026

**: on going activities up to 31st October 2026*

+39-3478463887

✉ stefano.giacomelli1992@gmail.com

✉ stefano.giacomelli@graduate.univaq.it

🐙 GitHub: StefanoGiacomelli

ORCID: 0009-0009-0438-1748

🔍 Google Scholar: Stefano Giacomelli

🌐 Personal Website: stefanogiacomelli.github.io

EDUCATION

- **Ph.D. XXXIX cycle in ICT - System Engineering, Telecommunications and HW/SW platforms** 2023 - 2026*
Dpt. of Information Engineering, Computer Science, and Mathematics, University of L'Aquila, Italy
Theme: Computational Methods for Auditory Scene Analysis and Synthesis supporting Immersive and Connected eXtended Reality Services, **Tutor:** Prof. Ch. F. Graziosi, **Co-Supervisor:** Prof. C. Rinaldi
- **M.A. in Sound Technology (DCSL34)** 2020 - 2023
Dpt. of Electronic Music, Conservatory of Music "A. Casella" of L'Aquila, Italy 110/110 cum laude w. honors
Thesis: "Vowel phonemes Analysis & Classification by means of OCON rectifiers Deep Learning architectures", **Supervisor:** Prof. M. Giordano
- **B.A. in Electroacoustic Music Composition (DCPL34)** 2015 - 2020
Dpt. of Music & New Technologies, Conservatory of Music "A. Casella" of L'Aquila, Italy 110/110 cum laude
Thesis: "...dans l'Espace" (for live electronics & octophonic feedback system), **Supervisor:** M° A. Di Scipio

RESEARCH ACTIVITIES

- **Invited Visiting Student – I3S Laboratory, Sophia Antipolis** Jul 2026–Oct 2026*
Visiting Ph.D. Researcher Université Côte d'Azur
 - Research on the reverse engineering of neural guitar-amplifier models into lightweight parametric Digital Signal Processing (DSP) structures for real-time audio effects. The activity investigates whether Neural Amp Modeler (NAM) profiles, represented through input-output transfer functions behavior and Impulse Response (IR) characteristics, can be approximated by controllable amplifier models implemented in FAUST and Web Audio Modules (WAM). The problem is formulated as a nonlinear system identification and parameter optimization, where a selected parametric model family is fitted to reproduce the response of a target neural profile. The considered model families cover multiple guitar tones and are implemented through modular stages such as nonlinear valve pre-amplification, tone-stack equalization, negative power-amplifier feedback, and speaker simulation. The experimental pipeline compares NAM-generated outputs with the outputs of candidate DSP models using audio-similarity objectives, perceptual or feature-based metrics, and short-segment optimization procedures. Candidate optimization strategies include Covariance Matrix Adaptation Evolution Strategy (CMA-ES), Nelder–Mead search, Bayesian optimization, and grid-based approaches, with additional attention to robustness, controllability, and real-time code feasibility. The broader objective is to assess whether *neural audio effects* can be compressed into interpretable, parametric, and computationally efficient real-time DSP models without losing the perceptual identity of the original system profile.
 - **Funded under** Ministerial Decree No. 118/2023, Mission 4, Component 1, Investment 4.1 of the National Recovery and Resilience Plan (PNRR) – “PNRR Research” – CUP E11I2300010000
 - **Host Supervisor:** Prof. M. Buffa (I3S - UCA)
 - **Official Website:** <https://www.i3s.unice.fr/en/>
- **Invited Visiting Student – Departement Elektrotechniek (ESAT-STADIUS)** Mar. 2026–May 2026
Visiting Ph.D. Researcher – Audio Engineering Laboratory KU Leuven

- Research on the adaptation of General-Purpose Audio Tagging (GP-AT) representations to spatially grounded Sound Event Localization and Detection (SELD). The activity addresses the structural mismatch between clip-level semantic audio inference and frame-level, multi-source, spatially explicit acoustic-scene trajectories. It combines a structured review of spatial audio representation learning with the design and evaluation of a Neural Architecture Search (NAS) framework for adapting pretrained GP-AT backbones to SELD tasks. The methodological study covers joint Sound Event Detection (SED) and Direction-of-Arrival (DOA) estimation, class-wise and track-wise formulations, Activity-Coupled Cartesian DOA estimation, Permutation-Invariant Training (PIT), convolutional-recurrent systems, ResNet–Conformer variants, attention-based temporal modeling, permutation-aware source tracking, and geometry-aware localization techniques. The experimental pipeline investigates model parameterization, activity-localization coupling, source-track assignment, targeted supervision, automatic knowledge distillation, and staged informing across SELD benchmark datasets, including STARSS23, TAU-NIGENS2021, TAU-NIGENS2020, and TAU2019. Diagnostic analyses focus on class imbalance, SED loss calibration, threshold sensitivity, active-only DOA supervision, dataset-dependent generalization, and interactions between semantic and spatial computational branches, with the objective of identifying when and how transferable semantic representations can support spatial *machine listening*.
- **Funded under** Ministerial Decree No. 118/2023, Mission 4, Component 1, Investment 4.1 of the National Recovery and Resilience Plan (PNRR) – “PNRR Research” – CUP E11I2300010000
- **Host Supervisor:** Prof. T. van Waterschoot (ESAT - KU Leuven)
- **Official Website:** <https://www.esat.kuleuven.be/stadius/>

• **Ph.D. in ICT – Computational Methods for ASA Supporting Immersive Extended Reality Services**

2023–2026*

Ph.D. Researcher (XXXIX Cycle)

University of L’Aquila

- Ph.D. research in Computational Auditory Scene Analysis (CASA), focused on computational methods for the analysis and synthesis of acoustic scenes in eXtended-Reality (XR) scenarios. The project treats acoustic-scene models both as recognition systems and as representational tools for describing sonic environments, spatial listening conditions, and situated auditory experiences. The research trajectory moves from human-inspired audio front-ends and differentiable perceptual operators to GP-AT models, considered as pragmatic implementations of transferable audio representation learning. These models are evaluated with respect to computational profile, analytical behavior, downstream task adaptation, cross-domain and inter-domain transfer, and knowledge-distillation capability. The work further examines the limits of current neural audio models and dataset organizations through semantic taxonomy analysis, domain-transfer experiments, and weakly controlled metadata or content structures, using these findings to motivate task-specialization tools and future directions toward *foundational reasoning* in automatic audio modeling. Methodological developments include Deep Learning (DL) pipelines based on lightweight convolutional-recurrent architectures, attention-based embedding layers, One-Class-One-Network (OCON) strategies, Kolmogorov–Arnold Networks (KANs) for eXplainable recognition, and task-specialized classification or tracking heads for Acoustic Scene Classification (ASC), Audio Tagging (AT), SED, SELD, and related scene-analysis tasks. Complementary research explores decomposition-based and auditory-modeling methods, including Empirical Mode Decomposition (EMD), Iterative Filtering (IF), neural approximations of iterative decomposition of non-stationary signals, and DSP-oriented front-end interfaces for source separation, event characterization, semantic processing, and localization-oriented applications. The activity also includes technical profiling, lightweight deployment analysis, reusable open-source implementation, and reproducible benchmarking of modular audio-AI components, with emphasis on the trade-off between task performance, computational cost, interpretability, and deployability in edge-oriented and resource-constrained settings.
- **Funded under** Ministerial Decree No. 118/2023, Mission 4, Component 1, Investment 4.1 of the National Recovery and Resilience Plan (PNRR) – “PNRR Research” – CUP E11I2300010000
- **Tutor:** Prof. Ch. F. Graziosi (DISIM - UnivAQ), **Co-Supervisor:** C. Rinaldi (CNIT)
- **Official Website:** <http://phdict.disim.univaq.it>

• **EAR – Enacting Artistic Research (WP3: Artificial and Collective Intelligence)**

2024–2026

AI Researcher & Developer

Conservatory of L’Aquila

- Research and development activity within *EAR – Enacting Artistic Research* Working Group 3, with responsibilities in the design of neural embedding models and multi-modal processing pipelines for artistic-research and technologically mediated musical archives. The contribution focuses on the reuse and adaptation of the pretrained EnCodec encoder from the Meta AudioCraft framework as the core audio representation module, together with a domain-oriented audio feature-processing pipeline for contemporary music, electroacoustic and electronic works, and 20th-century avant-garde repertoire. The pipeline integrates established audio-analysis libraries, including LibROSA and Essentia, with custom descriptors for acoustic, psychoacoustic, and musical-structural characterization, such as tonal-centroid representations, energy and onset descriptors, temporal grids, and other mid-level musical features. The activity also includes the design of a fine-tuning and retrieval-oriented infrastructure based on vector and graph databases, including ChromaDB and Neo4j, connected to a graphical interface for uploading and organizing multimedia content, including audio files, author metadata, textual documentation, and PDF attachments. A controlled and institutionally validated content-ingestion procedure was structured with the partner

institutions, also exploiting the dedicated Society for Artistic Research portal, to support the curated selection of materials for a domain-coherent semantic Retrieval-Augmented Generation (RAG) system. The broader objective is to develop an AI-assisted environment for the exploration, indexing, retrieval, and critical reflection of musico-logical, sonic-heritage, and technological-artistic research materials.

- **Funded by** the European Union – Next Generation EU, Mission 4, Component 1
- **References:** Prof. F. Ripa di Meana (ABA Roma), M° D. Macchione (ConsAQ)
- **Official Website:** <https://www.researchcatalogue.net/portals?portal=3258337>

• **Musical Metaverse: An Inclusive Extended Reality Platform for Networked Musical Interactions**

2024–2026

Technical Assistant

Conservatory of L'Aquila

- Technical and research-support activity within the PRIN 2022 project *Musical Metaverse*, coordinated by CIMI-Lab at the University of Trento with Politecnico di Torino, SAE Institute Milan, and the Conservatory of L'Aquila as partner institutions. The activity was carried out as an external technical and research collaborator for the Conservatory of L'Aquila, acting as a link between the technological outputs developed within the project and the testing, validation, and dissemination responsibilities assigned to the Conservatory. The contribution included the review and systematization of project research outputs, leading to a dedicated survey-oriented scientific contribution, as well as support for internal dissemination, demonstration, and presentation activities involving immersive headset-based access and simulated online front-end interaction. These activities supported parallel data collection on user experience, artistic and pedagogical usability, accessibility of immersive listening conditions, and future research requirements for Musical Metaverse services. The role also involved technical assistance in the evaluation of research products intended for composers, performers, and educational contexts, contributing to the scientific validation process from the perspective of musical practice, conservatory-based experimentation, and artistic-research assessment.
- **Funded by** the Italian Ministry of University and Research (MUR) and Next Generation EU – PRIN 2022, “Fondo per gli Investimenti nella Ricerca Scientifica e Tecnologica (FIRST)”, Chapter 7245, prot. n. 2022CZWWKP
- **References:** Prof. L. Turchet (CIMILab - UniTN), M° A. Di Scipio (MuEl - ConsAQ)
- **Official Website:** <https://musmet.eu>

• **MOVIDA – Manager-OVer-Ip-Detection-Audio**

2024–2025

AI Researcher & Developer

DEWS, University of L'Aquila – Tecnojest S.R.L.

- Industrial research and development activity within the collaboration between the Centre of Excellence DEWS at the University of L'Aquila and Tecnojest S.r.l., focused on deployable automatic audio analysis systems for smart-city monitoring and adaptive traffic-management infrastructures (MOVIDA). The contribution concerned the design, fine-tuning, and deployment of a convolutional-based sound-event detector for real-time recognition of emergency vehicle sirens. The system operates on spectrographic input mapped to Log-Mel representations and was adapted to the target detection task through dataset post-processing, architecture-level configuration, fine-tuning, and hyperparameter optimization. Deployment was oriented toward real-time and low-power inference on embedded hardware (RISC architectures), with implementation and research testing on Raspberry Pi 5 devices, with Linux headless operating system. The activity also included the development of a monitoring interface for signal acquisition and transmission, output logging, inference diagnostics, and runtime control of input-window management. This mechanism was used to verify the confidence of positive detections and reduce false positives during deployment, connecting neural audio detection with practical requirements for robust embedded and secure operation in noisy urban acoustic environments.
- **Funded by** internal DEWS–Tecnojest collaboration
- **References:** Prof. L. Pomante (DISIM - UnivAQ)
- **Official Websites:** <https://dews.univaq.it>, <https://www.invidea.it>

TEACHING ACTIVITIES

• **TelCom Engineering (DISIM), University of L'Aquila – Statistical Signal Processing & Multimedia**

2025

Guest Lecture: Listening to Architectures through DSP: Impulse Responses, Room Acoustics Descriptors, and Spatial Perception

- **Format:** Seminar lecture (2 hours) with slides, Python code and reference materials.
- **Abstract:** This seminar addressed Room Acoustics as a Digital Signal Processing (DSP) problem, interpreting enclosed spaces as linear time-invariant systems characterized through their sound Impulse Responses (IR). The lecture introduced the physical foundations of sound propagation in enclosed spaces, including air absorption, energy dissipation, and reverberant decay, and then discussed contemporary methodologies for Room Impulse Response (RIR) measurement, covering monaural, binaural, and Ambisonics-based approaches. Particular attention was devoted to practical DSP aspects such as noise floor estimation, dynamic range control, temporal windowing, and frequency-band analysis. The seminar then focused on the Energy Decay Curve (EDC) as the core numerical representation from which major room acoustics descriptors can be derived, illustrating the computation and interpretation of temporal, energetic, speech-related, and spatial parameters. Objective acoustic descriptors were presented in parallel with their perceptual implications, highlighting how the temporal and energetic structure of room responses shapes spatial listening experience, architectural sound perception, and the numerical representation of site-specific sonic environments.

– **Reference:** Prof. C. Rinaldi (CNIT), Prof. P. Di Marco (DISIM - UnivAQ)

• **Conservatory of Music "A. Casella" of L'Aquila – DREAM Ph.D. I Cycle**

2025

Seminar Series: Practical Fundamentals of Deep Learning for Audio Applications

- **Format:** 3-session seminar (9 hours) with slides, code references, and Colab IPython notebooks. Includes theoretical lectures and applied coding sessions.
- **Abstract:** This seminar provided a hands-on introduction to Deep Learning (DL) for audio, tailored for doctoral students in music and sound technology. Topics covered core concepts in Artificial Intelligence (AI), statistical Machine Learning (ML), and DL, with focus on Neural Networks foundational models: Perceptrons, Multi-Layer Perceptrons (MLPs), and Convolutional Neural Networks (CNNs), along with training pipelines for classification and regression. Particular attention was given to audio pre-processing, data augmentation, and the implementation of feature extraction pipelines using PyTorch-based audio frameworks. Sessions included guided exercises in Google Colab, fostering direct engagement with real-world codebases and experimental workflows. Later seminar modules addressed model design for artistic and analytical applications in music, including classification, signal analysis, and re-synthesis. The seminar aimed to bridge theoretical understanding and practical implementation for research-oriented projects in the sound and artistic domains.
- **Reference:** <https://www.consaq.it/offerta-formativa/phd/175-seminario-dottorale-dream.html>, M° A. Di Scipio (MuEl - ConsaQ)

• **TelCom Engineering (DISIM), University of L'Aquila – Statistical Signal Processing & Multimedia**

2024

Guest Lecture: Cepstral Features for Discrete Audio Signal Analysis

- **Format:** Seminar lecture (4 hours) with slides, Python code examples, reference materials, and final exam assignments.
- **Abstract:** This lecture introduced fundamental principles and practical applications of DSP for audio, with a specific focus on *Cepstral Analysis*. Key topics included the mathematical foundations and signal-domain interpretation of the cepstrum, quefrency, and liftering techniques. The lecture detailed the computational role of Cepstral Coefficients (CCs), covering filterbank design, logarithmic compression, and Discrete Cosine Transform (DCT)-based de-correlation. Special emphasis was placed on their use in speech and general audio classification tasks. Examples addressed time-frequency representations, pre-emphasis filtering, and windowing strategies, with a final discussion on feature dimensionality, perceptual scale modeling, and common pitfalls in practical deployments.
- **Reference:** Prof. C. Rinaldi (CNIT), Prof. P. Di Marco (DISIM - UnivAQ)

• **Thesis & Exams Co-Supervisor**

Department of Information Engineering, Computer Science, and Mathematics (DISIM), University of L'Aquila

- C. Palermo 2025, **Bachelor Degree Thesis** in Information Engineering (Control) - "*Design, Implementation, and Analysis of a VST Reverberator based on Gardner's Model*" (DISIM - UnivAQ)
- M. Angelucci 2024, **Bachelor Degree Thesis** in Information Engineering - "*Effects of Audio Scenarios on User Perception and Immersivity in Virtual Reality*" (DISIM - UnivAQ)
- R. Prosperococco 2024, **Final Assignment** for "Statistical Signal Processing & Multimedia" Master course - *Human Factor Cepstral Coefficients* with MATLAB implementations (DISIM - UnivAQ)

TECHNICAL SKILLS AND INTERESTS

Languages: Italian (*native*), English

Programming Languages: Python, MATLAB, FAUST, C++, HTML, CSS, JavaScript, Bash

Scientific packages & Machine Learning: NumPy, Numba, CuPy, Pandas, SciPy, Matplotlib, Seaborn, scikit-learn, PyTorch, PyTorch Lightning, TorchMetrics, TorchAudio, TorchVision, ONNX, pytest, Weights&Biases, TensorFlow, TensorBoard

Audio, Music, and Signal Processing: librosa, Essentia, soundfile, sounddevice, PyAudio, Mido, music21, DCASE_utils, DCASE_eval, SimPy, PyQt5

Databases, Version Control, and Infrastructure: Git, GitHub, GitLab, ChromaDB, Neo4j, Proxmox, VMware, AnyDesk, TeamViewer

IDEs and Development Environments: VSCode, PyCharm, Anaconda, Google Colab, Lightning AI Studio, Cursor, Google Antigravity, Sublime Text

Office, Writing, and Research Tools: Overleaf (LaTeX), Zotero, Markdown, Microsoft Office, Google Workspace

Audio/Video and Multimedia: REAPER, Ableton Live, Pro Tools, Max (Gen, RNBO, Jitter), Pure Data (PlugData), Audacity, Adobe Audition, Sonic Visualizer, TouchDesigner, DaVinci Resolve

Courseworks:

- **Bachelor:** Acoustics, Psychoacoustics, Perception Theories, Electroacoustics, Electroacoustic Music Composition & Analysis, Contemporary Music Analysis, Music Theory, Harmony, Composition, Orchestra & Ensemble Arrangement, Music History, Electroacoustic Music History, Multimedia, Live Performance, Music Computing

- **Master:** Advanced Music Computing, Architectural Acoustics, Spatial Sound, Sound Art & Installation, Advanced Live Performance, Contemporary Music Semiography, Algorithmic Composition, Electroacoustic Music History, Audio Restoration
- **Ph.D.:** Ecological Statistics, Statistical Signal Processing & Multimedia, Digital Signal Processing with Programmable Hardware, Neural Networks & Deep Learning (Ph.D. Advanced Course, Scuola Superiore Sant'Anna, Pisa), Mathematics for Signal Processing (Ph.D. Advanced Course, Department of Mathematics, University of L'Aquila), Summer School in Mathematics for Signal Processing and Applications in Geophysics and Other Fields (MaSAG 2025)

Areas of Interest: Sound/Audio, Acoustics (Room Acoustics), Soundscape studies, Sound ecology, Computational musicology, Electroacoustic and Electronic Music, Multimodal-audio retrieval, Computational Auditory Scene Analysis (CASA), Digital Signal Processing (DSP), Audiology, Audio Representation learning, General-Purpose Audio Tagging (GP-AT), Acoustic Scene Classification (ASC), Sound Event Detection (SED), Sound Event Localization and Detection (SELD), Sound Source Separation (SSS), Spatial audio, Knowledge Distillation (KD), embedded audio AI, Artistic Research, Optimization

SOCIETY AFFILIATIONS

- **IEEE** Graduate Student Member, R8-Europe Italy Section No.: 100317446
- **IEEE Societies:** Young Professionals, Communications Society, Computational Intelligence Society, Computer Society, Control Systems Society, Information Theory Society, Signal Processing Society, Cloud Computing Technical Community, Distributed Processing Technical Community, High Performance Technical Community, Robotics and Automation Technical Committee on Machine Learning for Automation, Intelligent Informatics Technical Community, Internet of Things Community, Multimedia Computing Technical Community, Parallel Processing Technical Community, Pattern Analysis & Machine Intelligence Technical Community, Real Time Systems Technical Community
- **EURASIP** European Association for Signal Processing
- **DCASE** Detection & Classification of Acoustic Scenes & Events Community
- **ISMIR** International Society for Music Information Retrieval
- **AIMI** Associazione Informatica Musicale Italiana
- **SAR** Society for Artistic Research
- **IoS** Internet of Sounds Research Network
- **Peer-Reviewer** for IEEE Transactions on Audio, Speech and Language Processing (TASLP), Springer Nature Journal of Audio Speech and Music Processing (JASMP), IEEE Symposium on Computers and Communications (ISCC), IEEE ISCC New Generation Multimedia Service at the Edge Workshop (NGMSE), IEEE International Symposium on the Internet of Sounds (IS²), Linux Audio Conference (LAC), Colloqui di Informatica Musicale (CIM)

ACHIEVEMENTS & AWARDS

- **XV Premio Nazionale delle Arti – 1st Place (Category D)**
W. Kotonski – "Etiuda na jedno uderzenie w talerz" (Study on One Cymbal Stroke)
 Real-time digital re-synthesis (with score-following)
Link: <https://musel.consaq.it/2021/04/24/pna-edizione-xv/> 2021
- **XV Premio Nazionale delle Arti – Honorable Mention (Category A)**
S. Giacomelli – "...dans L'Espace (inside my bedroom)"
 Octophonic acousmatic work (live sound feedback performance)
Link: <https://musel.consaq.it/2021/04/24/pna-edizione-xv/> 2021
- **XIV Premio Nazionale delle Arti – 2nd Place (Category B)**
Elec3 Laptop Trio (S. Giacomelli, D. Scorrane, L. Canzonetti) – "Portrait.Elec3"
 Quadraphonic live performance (real-time soundscape processing & digital feedback)
Link: <https://www.consaq.it/2019/3257/70/pna-edizione-xiv/> 2019
- **XIII Premio Nazionale delle Arti – 1st Place (Category B) - Sound Director**
M. Di Berardino – "Trasfigurazioni Materiche"
 Quadraphonic live performance (for basque tambourine and real-time granulation)
Link: <https://musel.consaq.it/2018/10/06/xiii-premio-nazionale-delle-arti-musica-elettronica-selezioni-finali/> 2018

Journal Articles

- [1] Stefano Giacomelli et al. “AudioSet-Tools: A Python Framework for Taxonomy-Aware AudioSet Curation and Reproducible Audio Research”. In: *EURASIP Journal of Audio, Speech and Music Processing* 2026.2 (2026). DOI: 10.1186/s13636-025-00436-z.
- [2] Stefano Giacomelli et al. “From General-Purpose Audio Tagging to Real-Time Emergency Vehicle Siren Detection”. In: *IEEE Transactions on Audio, Speech and Language Processing* (2026), pp. 1–16. DOI: 10.1109/TASLPRO.2026.3703169.

Conference Proceedings

- [3] Stefano Giacomelli, Marco Giordano, and Claudia Rinaldi. “The OCON model: an Old but Gold Solution for Distributable Supervised Classification”. In: *2024 IEEE Symposium on Computers and Communications (ISCC)*. Los Alamitos, CA, USA: IEEE Computer Society, 2024, pp. 1–7. DOI: 10.1109/ISCC61673.2024.10733621. URL: <https://doi.ieeecomputersociety.org/10.1109/ISCC61673.2024.10733621>.
- [4] Stefano Giacomelli, Marco Giordano, and Claudia Rinaldi. “The OCON Model: An Old but Green Solution for Distributable Supervised Classification for Acoustic Monitoring in Smart Cities”. In: *2024 IEEE 5th International Symposium on the Internet of Sounds (IS2)*. 2024, pp. 1–10. DOI: 10.1109/IS262782.2024.10704155.
- [5] Stefano Giacomelli et al. “A Python Tool for Time-Varying RIR Dataset Generation”. In: *2024 International Workshop on Sound Signal Processing Applications (IWSSPA)*. 2024, pp. 1–10. DOI: 10.5281/zenodo.14193309.
- [6] Stefano Giacomelli et al. “Remote Immersive Audio Production: State of the Art Implementation, Challenges, and Improvements”. In: *2024 IEEE 5th International Symposium on the Internet of Sounds (IS2)*. 2024, pp. 1–10. DOI: 10.1109/IS262782.2024.10704192.
- [7] Massimiliano Cerioni et al. “Giving AI ears: a Perceptual Approach to AI For Artistic Research”. In: *poster presented at 16th International Conference on Artistic Research*. University of Porto, PO, Portugal: Society for Artistic Research (SAR), 2025. URL: <https://www.researchcatalogue.net/view/2759372/3582572>.
- [8] Alberto Maria Gatti et al. “DawBi: A WebSocket-Based Plugin for Semantic Dialogue Between DAW and KOBi AI”. In: *2025 9th Web Audio Conference (WAC) IRCAM - Mozilla*. 2025, pp. 1–10. DOI: 10.5281/zenodo.17750005.
- [9] Stefano Giacomelli, Agostino Di Scipio, and Claudia Rinaldi. “The Musical Metaverse Research Project: a MusicTech Survey”. In: *2025 IEEE 6th International Symposium on the Internet of Sounds (IS2), 2nd International Workshop on Musical Metaverse*. 2025, pp. 1–10. DOI: 10.1109/IS264627.2025.11284663.
- [10] Stefano Giacomelli et al. “Design and Deployment of a Standard Framework for Audio Neural Networks Embedding Models”. In: *2025 IEEE Symposium on Computers and Communications (ISCC), New Generation Multimedia Services at the Edge Workshop (NGMSE)*. Bologna, Italy, 2025. DOI: 10.1109/ISCC65549.2025.11326439.
- [11] Marco Giordano et al. “Real-Time Emergency Vehicle Siren Detection with Efficient CNNs on Embedded Hardware”. In: *2025 IEEE 6th International Symposium on the Internet of Sounds (IS2)*. 2025, pp. 1–10. DOI: 10.1109/IS264627.2025.11284671.
- [12] Marco Pennese, Stefano Giacomelli, and Claudia Rinaldi. “A Kolmogorov Arnold Network NAS Framework for Strumming Pattern Recognition in Technology-Enhanced Pop/Rock Music Education”. In: *2025 IEEE 6th International Symposium on the Internet of Sounds (IS2)*. 2025, pp. 1–10. DOI: 10.1109/IS264627.2025.11284580.

- [13] Claudia Rinaldi et al. "Beyond Marketing: a Holistic Analysis of 3D Audio Workflows for Live Music Production". In: *2025 IEEE 6th International Symposium on the Internet of Sounds (IS2), 3rd International Workshop on Networked Immersive Audio*. 2025, pp. 1–10. DOI: 10.1109/IS264627.2025.11284654.
- [14] Stefano Giacomelli et al. "Torch AMT: Differentiable Auditory Modeling for PyTorch-based Edge Audio Systems". In: *2026 IEEE Symposium on Computers and Communications (ISCC), New Generation Multimedia Services at the Edge Workshop (NGMSE)*. Villamoura, Algarve, Portugal, 2026. DOI: TBD.
- [15] Marco Giordano et al. "DASSS and KineScaper: Open Python Frameworks for Physically-Grounded Synthetic Emergency Vehicle Siren Dataset Generation". In: *2026 IEEE Symposium on Computers and Communications (ISCC), New Generation Multimedia Services at the Edge Workshop (NGMSE)*. Villamoura, Algarve, Portugal, 2026. DOI: TBD.

Datasets, Preprints & Technical Reports

- [16] Stefano Giacomelli and Claudia Rinaldi. *AudioSet-EV: an AudioSet-derived Distribution of Emergency Vehicle Siren Sounds*. 2025. DOI: 10.5281/ZENODO.14882314. URL: <https://zenodo.org/doi/10.5281/zenodo.14882314>.
- [17] Stefano Giacomelli et al. *From Large-scale Audio Tagging to Real-Time Explainable Emergency Vehicle Sirens Detection*. 2025. DOI: 10.48550/ARXIV.2506.23437. URL: <https://arxiv.org/abs/2506.23437>.
- [18] Marco Pennese, Stefano Giacomelli, and Claudia Rinaldi. *The Strummin' Dataset: an international pop/rock curated audio selection for strumming patterns recognition*. Version 1.0. July 2025. DOI: 10.5281/zenodo.15862786. URL: <https://doi.org/10.5281/zenodo.15862786>.
- [19] Stefano Giacomelli and Claudia Rinaldi. *AudioSet-EV v2: a refined AudioSet-derived Distribution of Emergency Vehicle Siren Sounds*. Version v2.0. Feb. 2026. DOI: 10.5281/zenodo.18668076. URL: <https://zenodo.org/uploads/18668076>.
- [20] Marco Giordano, Stefano Giacomelli, and Claudia Rinaldi. *KineScaper-EV: A Controlled Dataset for Emergency Vehicle Siren Recognition, Siren-Type Benchmarking, and Detection-Oriented Analysis*. Version v1.0. 2026. DOI: 10.5281/zenodo.19163980. URL: <https://zenodo.org/uploads/19163980>.

ARTISTIC PRODUCTIONS & CONCERTS

Where not otherwise specified, the role is Sound Director and Audio Technician.

- **Notte della Ricerca e dei Ricercatori 2025 (Première)**
Tessela - Audio/Video Installation (EAR WP3 - Audio Development DEMO)
 Ex-Mattatoio, Campo Boario (Accademia di Belle Arti di Roma - ABA Roma) 2022
- **Concerti di Studio di Primavera (VI ed.)**
N. Collins - Pea Soup (S. Giacomelli - live electronics, C. Rinaldi - trumpet)
 Auditorium "E. Morricone" (Università degli studi di Roma - Tor Vergata) 2022
- **EMUFestSABINA 2022**
G. Boccio - Percorsi Armonici (S. Giacomelli - live electronics, G. Boccio - cello)
B. Truax - Nightwatch (A. Gizzi - marimba)
W. Kotonski - Etiuda na jedno uderzenie w talerz
 Teatro Comunale, Vicolo Primo, Poggio Moiano (Rieti) 2022
- **VII rassegna ElettroAQuistica**
B. Truax - Nightwatch (A. Gizzi - marimba)
 Auditorium "Shigeru Ban" (Conservatorio di Musica "A. Casella" di L'Aquila) 2022
- **MAXXI L'Aquila - ElettroAQuistica Ensemble**
B. Truax - Nightwatch (A. Gizzi - marimba)
A. Lucier - Music for pure waves, bass drums and acoustic pendulums (A. Gizzi - live electronics)
A. Di Scipio - Ecosistemico Udibile n.1 (A. Di Scipio - live electronics, S. Giacomelli - clave)
 Museo Nazionale delle arti del XXI secolo (MAXXI, L'Aquila) 2022

- **Stagione Musicale Comunale di Bisenti - I Suoni Narranti**
L. Bellini, F. Di Pace - "MusiCola: Leggende del Mediterraneo" (Orchestra "A. Casella")
 Bisenti (Teramo) 2022
- **L'organo nel Tempo - Nuova Consonanza (Première)**
S. Giacomelli - "...dans L'Espace Orgue" (S. Giacomelli - live electronics, F. Gianfreda - orgue)
 Chiesa di S. Giorgio (Rieti) 2022
- **ArteScienza 2022 (Première)**
S. Giacomelli - "...dans L'Espace" interactive installation
 Goethe Institut Rom, Aula 114 (Roma) 2022
- **Musei Capitolini - Musica in Mostra: Contrapposizioni e Convergenze**
S. Giacomelli - "...dans L'Espace"
 Musei Capitolini di Castel Sant'Angelo (Roma) 2022
- **EMUFestSABINA 2021**
C. Rinaldi - "Ti Prendo per Mano" (S. Giacomelli - live electronics, C. Rinaldi - trumpet)
N. Collins - "Pea Soup" (S. Giacomelli - live electronics, M. Pennese: trombone, C. Rinaldi: trumpet)
M. Pennese - "Or-mai" (M. Pennese - guitar) (Première)
S. Giacomelli - "...dans l'Espace (inside my bedroom)"
 Teatro Comunale Vicolo Primo, Poggio Moiano (Rieti) 2021
- **EMUFestSABINA - Festa della Musica 2021**
N. Collins - "Pea Soup" (S. Giacomelli - live electronics, M. Pennese - trombone, C. Rinaldi - trumpet)
 Chiesa di San Rufo (Rieti) 2021
- **ArteScienza 2021**
S. Giacomelli - "...dans L'Espace Acousmatique"
 Goethe Institut Rom, Giardino Basso (Roma) 2021
- **EMUFestSABINA 2020 (Première)**
S. Giacomelli - "...dans L'Espace"
 Chiesa di S. Martino, Poggio Moiano (Rieti) 2020
- **IV rassegna ElettroAqustica (Première)**
G. Scurti - "Ricerca su un Respiro Sospeso" (S. Giacomelli - live electronics, G. Scurti - bayan)
 Auditorium "Shigeru Ban", Conservatorio di Musica "A. Casella" di L'Aquila 2018
- **Rassegna "Micro-Fonie - Le Forme del Suono: Musiche della Contemporaneità"**
K. Stockhausen - "Mikrophonie I" (S. Giacomelli - IT developer, tam)
 Auditorium "R. Caetani", Conservatorio di Musica "O. Respighi" di Latina 2018
- **XXII Colloquio di Informatica Musicale (Première)**
S. Giacomelli - "I_Side #2 (flow of sensations)" (M. Di Berardino - sound director)
 Sala "G. Verdi", Conservatorio di Musica "J. Tomadini" di Udine 2018
- **III rassegna ElettroAqustica (Première)**
G. Boccio - "Percorsi Armonici" (S. Giacomelli - sound director / live electronics, G. Boccio - cello)
 Auditorium "Shigeru Ban", Conservatorio di Musica "A. Casella" di L'Aquila 2017
- **I Cantieri dell'Immaginario (Società B. Barattelli, VI ed.)**
K. Stockhausen - "Mikrophonie I" (S. Giacomelli - IT developer, tam)
 Museo Nazionale d'Abruzzo (MUNDA, L'Aquila) 2017
- **Scarlatti Lab Electronics (Società Scarlatti Contemporanea)**
K. Stockhausen - "Mikrophonie I" (S. Giacomelli - IT developer, tam)
 Domus Ars (Napoli) 2017